

Nonconvex Instability of Fisher–Rao Gaussian Discretizations and Running-Minimum Bures–Wasserstein Stationarity

Abstract

We construct deterministic one-dimensional Gaussian experiments for the smooth non-log-concave target

$$V_R(x) = \frac{1}{2}x^2 - 2R^2 \log \cosh(x/R), \quad -1 \leq V_R''(x) \leq 1.$$

Although the Hessian is globally bounded, the two Gaussian Fisher–Rao covariance discretizations can generate very large trajectory-wise Fisher–Rao gradient norms. The Riemannian exponential covariance step undergoes a negative-curvature cascade from $c_0 = 1$, while the KL/Bregman covariance step has a pole near the loss of positive definiteness. In contrast, the Bures–Wasserstein forward–backward step is evaluated through the running-minimum Bures–Wasserstein gradient norm, the quantity appearing in the nonconvex stationarity theory. The experiment is numerical evidence for the absence of a smoothness-only, covariance-free, trajectory-wise Fisher–Rao gradient control; it is not a literal mathematical proof from finite numerics.

1 Setup

For $X \sim \mathcal{N}(m, c)$ define

$$A_R(m, c) = \mathbb{E}[V_R''(X)], \quad \mathcal{F}(m, c) = \mathbb{E}[V_R(X)] - \frac{1}{2} \log c.$$

The experiments are deterministic and CPU-only. All expectations are evaluated by fixed Gauss–Hermite quadrature; no normalization constant for the target is needed. The derivatives are implemented as

$$V_R'(x) = x - 2R \tanh(x/R), \quad V_R''(x) = 1 - 2\{1 - \tanh^2(x/R)\}.$$

The symmetric trajectory experiments use $m = 0$, so we write $A_R(c) = A_R(0, c)$. The target is normalizable and non-log-concave: $V_R''(0) = -1$, while $V_R''(x) \rightarrow 1$ in the tails.

2 Algorithms

The Riemannian Fisher–Rao covariance discretization is

$$c_{n+1} = c_n \exp\{\Delta t(1 - c_n A_R(c_n))\}.$$

The implementation evolves $\log c_n$ and stops when $c_n > R^2/4$ or when the log-covariance overflow guard is reached.

The KL/Bregman Fisher–Rao covariance discretization is

$$c_{n+1} = \frac{(1 + \Delta t)c_n}{1 + \Delta t c_n A_R(c_n)}.$$

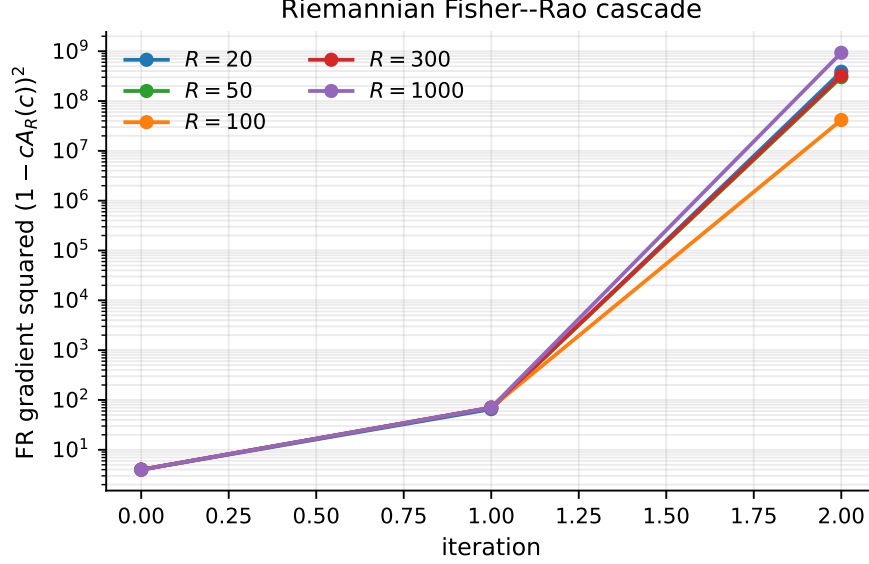


Figure 1: Riemannian Fisher–Rao covariance cascade from $c_0 = 1$, $\Delta t = 1$. The trajectory-wise Fisher–Rao gradient squared grows rapidly despite the global Hessian bound $|V_R''| \leq 1$.

If the denominator is non-positive, the step is marked non-SPD.

The Bures–Wasserstein forward–backward step is

$$\tilde{c}_n = (1 - \eta A_R(c_n))^2 c_n, \quad c_{n+1} = \frac{1}{2} \{ \tilde{c}_n + 2\eta + \sqrt{\tilde{c}_n(\tilde{c}_n + 4\eta)} \}.$$

We record two stationarity diagnostics,

$$\|\nabla_{\text{FR}} \mathcal{F}\|^2 = (1 - cA_R(c))^2, \quad \|\nabla_{\text{BW}} \mathcal{F}\|^2 = \frac{(1 - cA_R(c))^2}{c}.$$

For the BW experiment the comparison is to $\min_{0 \leq k < N} \|\nabla_{\text{BW}} \mathcal{F}(c_k)\|^2$, not to the pointwise Fisher–Rao norm.

3 Results

Figure 1 shows the Riemannian cascade. Near $c = 1$ and large R , the Gaussian samples see almost constant negative curvature $A_R(c) \approx -1$, so the multiplicative factor is approximately $\exp(1 + c)$. This produces a covariance burst and a large value of $(1 - cA_R(c))^2$ before the stopping criterion is reached.

Figure 2 isolates the rational pole in the KL/Bregman scheme. With $c_0 = 1 - \varepsilon$ and $A_R(c_0) \approx -1$, the denominator is close to ε and c_1 is close to $2/\varepsilon$. The plotted Fisher–Rao gradient at the next state follows the pole trend until the finite- R curvature average no longer behaves like a constant -1 over the entire Gaussian cloud.

Figure 3 uses the stationarity quantity controlled by the nonconvex BW forward–backward theory. The envelope is intentionally loose, because Δ scales with the target depth. The important comparison is geometric: the BW claim is a running-minimum BW-gradient statement, whereas the Fisher–Rao instability figures are trajectory-wise FR-gradient diagnostics.

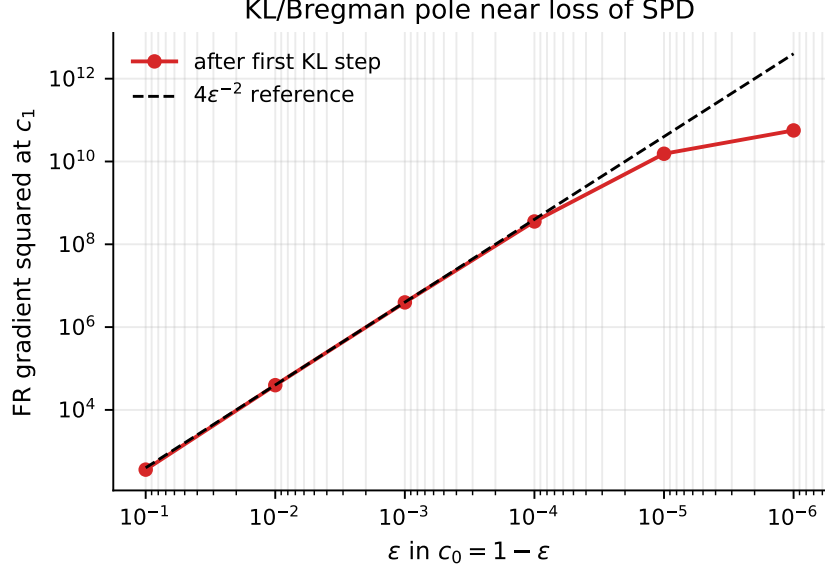


Figure 2: KL/Bregman pole experiment at $R = 1000$, $\Delta t = 1$, $c_0 = 1 - \epsilon$. The reference line shows the formal $4\epsilon^{-2}$ scaling associated with the pole approximation.

4 Findings and Implications

These experiments show that a globally bounded Hessian alone does not provide a useful covariance-free trajectory-wise control of the Gaussian Fisher–Rao gradient norm for these discretizations. The obstruction is not roughness of the target: V_R is smooth and has $-1 \leq V_R'' \leq 1$. Instead, negative curvature couples to covariance through the factor $cA_R(c)$, and both Fisher–Rao discretizations can amplify that factor sharply.

The BW comparison should not be read as a pointwise Fisher–Rao guarantee. It is a different geometry and a different nonconvex conclusion: the running minimum of the BW gradient norm is compared with an $O(1/N)$ envelope.

5 Limitations

The evidence is finite-dimensional, finite-precision, and finite-horizon. It supports the claimed absence of a smoothness-only covariance-free trajectory-wise Fisher–Rao bound, but it is not a formal impossibility theorem. The Gauss–Hermite quadrature order, overflow guards, and numerical optimizer for $\mathcal{F}_\star^{\text{num}}$ are saved with the output metadata to make the experiment reproducible.

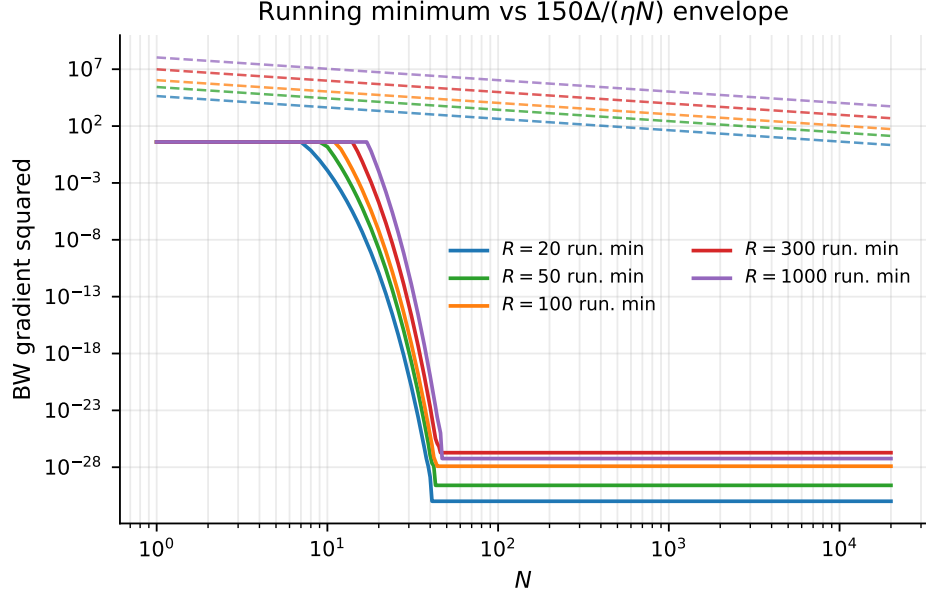


Figure 3: Bures–Wasserstein forward–backward runs with $\eta = 0.9$. Solid curves are the running minimum of the BW gradient squared; dashed curves are the loose envelope $150\Delta/(\eta N)$ with $\Delta = \mathcal{F}(c_0) - \mathcal{F}_\star^{\text{num}}$.

experiment	R	max / slope	terminal value	check
Riemannian cascade	20	2×10^4	2.4×10^4	c_threshold
Riemannian cascade	50	1.7×10^4	3.1×10^4	c_threshold
Riemannian cascade	100	6.4×10^3	3.2×10^4	c_threshold
Riemannian cascade	300	1.8×10^4	3.2×10^4	c_threshold
Riemannian cascade	1000	3.1×10^4	3.2×10^4	log_c_overflow_guard
KL pole	1000	-1.998	5.6×10^{10}	pole-window slope
BW forward–backward	20	1×10^{-31}	2.17	yes
BW forward–backward	50	2.6×10^{-30}	13.6	yes
BW forward–backward	100	1.2×10^{-28}	54.4	yes
BW forward–backward	300	1.9×10^{-27}	490	yes
BW forward–backward	1000	5.8×10^{-28}	5.4×10^3	yes

Table 1: Summary diagnostics. Riemannian rows report the maximum Fisher–Rao gradient norm along each cascade; the KL row reports the log–log slope of the first-step pole diagnostic; BW rows report the final running minimum and final theorem envelope check.